

assignment 7

2023-03-06

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

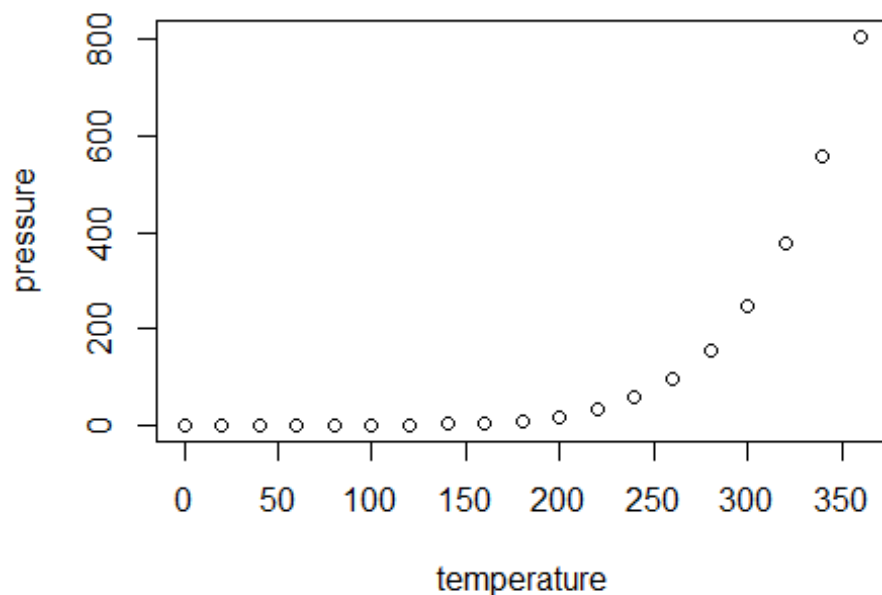
When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

```
##      speed      dist
##  Min.   : 4.0    Min.   :  2.00
## 1st Qu.:12.0    1st Qu.: 26.00
## Median :15.0    Median : 36.00
## Mean   :15.4    Mean   : 42.98
## 3rd Qu.:19.0    3rd Qu.: 56.00
## Max.   :25.0    Max.   :120.00
```

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

```
library(tidyverse)
```

```
## Attaching packages: tidyverse
1.3.2
## ✓ ggplot2 3.4.0      ✓ purrr 1.0.1
## ✓ tibble 3.1.8       ✓ dplyr 1.0.10
## ✓ tidyr 1.3.0        ✓ stringr 1.5.0
## ✓ readr 2.1.3        ✓ forcats 0.5.2
## Conflicts:
tidyverse_conflicts()
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag() masks stats::lag()
```

```
library(openintro)
```

```
## Loading required package: airports
## Loading required package: cherryblossom
## Loading required package: usdata
```

```
library(statsr)
```

```
## Loading required package: BayesFactor
## Loading required package: coda
## Loading required package: Matrix
##
```

```
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
## *****
## Welcome to BayesFactor 0.9.12-4.4. If you have questions, please contact
## Richard Morey (richarddmorey@gmail.com).
##
## Type BFManual() to open the manual.
## *****
##
## Attaching package: 'statsr'
##
## The following objects are masked from 'package:openintro':
##
##     calc_streak, evals, nycflights, present
```

```
library(broom)
```

```
##What are the dimensions of the dataset? What does each row represent?
```

```
dim(hfi)
```

```
## [1] 1458 123
```

```
##The dataset spans a lot of years, but we are only interested in data from year 2016. Filter
the data hfi data frame for year 2016, select the six variables, and assign the result to a data
frame named hfi_2016.
```

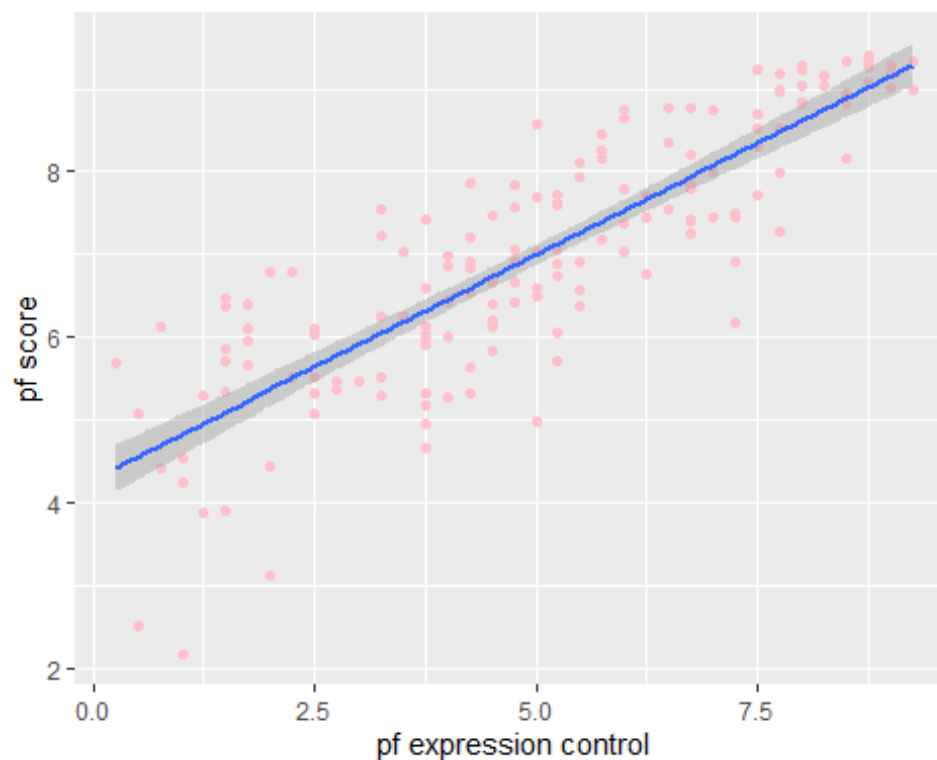
```
hfi_2016 <- hfi %>%
  filter(year == 2016) %>%
  select(year, hf_score, pf_score, pf_expression_control)
hfi_2016
```

```
## # A tibble: 162 × 4
##   year hf_score pf_score pf_expression_control
##   <dbl>   <dbl>   <dbl>           <dbl>
## 1  2016     7.57     7.60             5.25
## 2  2016     5.14     5.28             4
## 3  2016     5.64     6.11             2.5
## 4  2016     6.47     8.10             5.5
## 5  2016     7.24     6.91             4.25
## 6  2016     8.58     9.18             7.75
## 7  2016     8.41     9.25             8
## 8  2016     6.08     5.68             0.25
## 9  2016     7.40     7.45             7.25
## 10 2016     6.85     6.14             0.75
## # ... with 152 more rows
```

What type of plot would you use to display the relationship between the personal freedom score, `pf_score`, and `pf_expression_control`? Plot this relationship using the variable `pf_expression_control` as the predictor. Does the relationship look linear? If you knew a country's `pf_expression_control`, or its score out of 10, with 0 being the most, of political pressures and controls on media content, would you be comfortable using a linear model to predict the personal freedom score?

```
ggplot(hfi_2016, aes(pf_expression_control, pf_score)) +
  geom_point(color = "pink") +
  geom_smooth(method = 'lm', se = TRUE) +
  xlab("pf expression control") +
  ylab("pf score")
```

`'geom_smooth()'` using formula = `'y ~ x'`



```
hfi_2016 %>%
  summarise(cor(pf_expression_control, pf_score))
```

```
## # A tibble: 1 × 1
##   `cor(pf_expression_control, pf_score)`
##                                     <dbl>
## 1                                     0.845
```

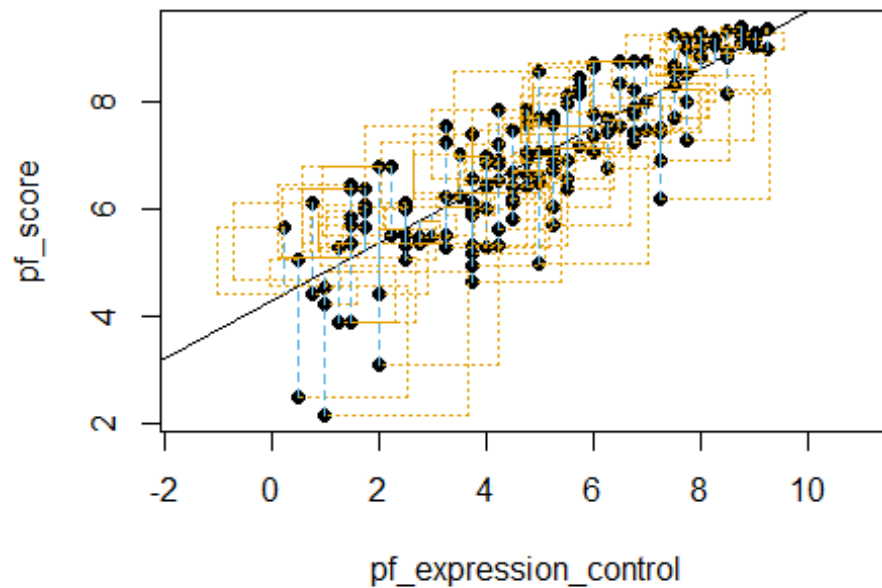
this plot looks linear and i would use this model to pridict corrolation coff is 0.84

##Q: Looking at your plot from the previous exercise, describe the relationship between these two variables. Make sure to discuss the form, direction, and strength of the

relationship as well as any unusual observations.

##The correlation is positive and strong. As the pf control expression increases the pf score increases aswell.

```
plot_ss(x = pf_expression_control, y = pf_score, data = hfi_2016, showSquares = TRUE)
```



```
## Click two points to make a line.
```

```
## Call:
```

```
## lm(formula = y ~ x, data = pts)
```

```
##
```

```
## Coefficients:
```

```
## (Intercept)          x
```

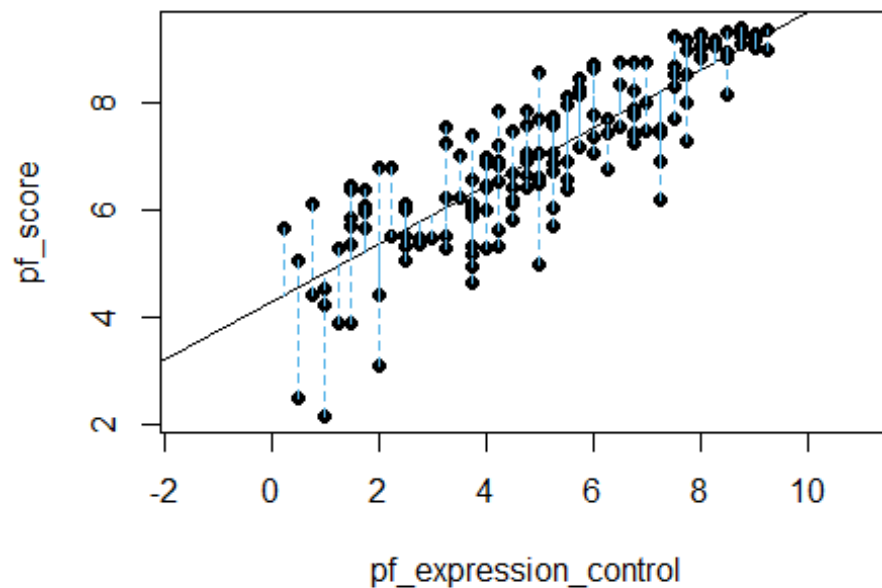
```
##      4.2838      0.5418
```

```
##
```

```
## Sum of Squares: 102.213
```

##Using plot_ss, choose a line that does a good job of minimizing the sum of squares. Run the function several times. What was the smallest sum of squares that you got? How does it compare to your neighbours?

```
plot_ss(x = pf_expression_control, y = pf_score, data = hfi_2016)
```



```
## Click two points to make a line.
```

```
## Call:
```

```
## lm(formula = y ~ x, data = pts)
```

```
##
```

```
## Coefficients:
```

```
## (Intercept)          x
```

```
##      4.2838      0.5418
```

```
##
```

```
## Sum of Squares: 102.213
```

```
##Linear model
```

```
m1 <- lm(pf_score ~ pf_expression_control, data = hfi_2016)
```

```
tidy(m1)
```

```
## # A tibble: 2 × 5
```

```
##   term                estimate std.error statistic  p.value
```

```
##   <chr>                <dbl>     <dbl>     <dbl>    <dbl>
```

```
## 1 (Intercept)         4.28      0.149      28.8 4.23e-65
```

```
## 2 pf_expression_control 0.542    0.0271     20.0 2.31e-45
```

```
glance(m1)
```

```
## # A tibble: 1 × 12
```

```
##   r.squared adj.r.squa...1 sigma stati...2 p.value    df logLik   AIC   BIC
```

```
devia...3
```

```
##   <dbl>         <dbl> <dbl>    <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl>
```

```
<dbl>
## 1      0.714      0.712 0.799    400. 2.31e-45      1 -193. 391. 400.
102.
## # ... with 2 more variables: df.residual <int>, nobs <int>, and abbreviated
## # variable names ^adj.r.squared, ^statistic, ^deviance
```

Fit a new model that uses `pf_expression_control` to predict `hf_score`, or the total human freedom score. Using the estimates from the R output, write the equation of the regression line. What does the slope tell us in the context of the relationship between human freedom and the amount of political pressure on media content?

```
table22 <- lm(hf_score ~ pf_expression_control, data = hfi_2016)
summary(table22)
```

```
##
## Call:
## lm(formula = hf_score ~ pf_expression_control, data = hfi_2016)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.68164 -0.45467  0.05692  0.46699  1.88128
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      5.05340    0.12293   41.11  <2e-16 ***
## pf_expression_control  0.36843    0.02236   16.48  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6595 on 160 degrees of freedom
## Multiple R-squared:  0.6291, Adjusted R-squared:  0.6268
## F-statistic: 271.4 on 1 and 160 DF, p-value: < 2.2e-16
```

If someone saw the least squares regression line and not the actual data, how would they predict a country's personal freedom score for one with a 3 rating for `pf_expression_control`? Is this an overestimate or an underestimate, and by how much? In other words, what is the residual for this prediction?

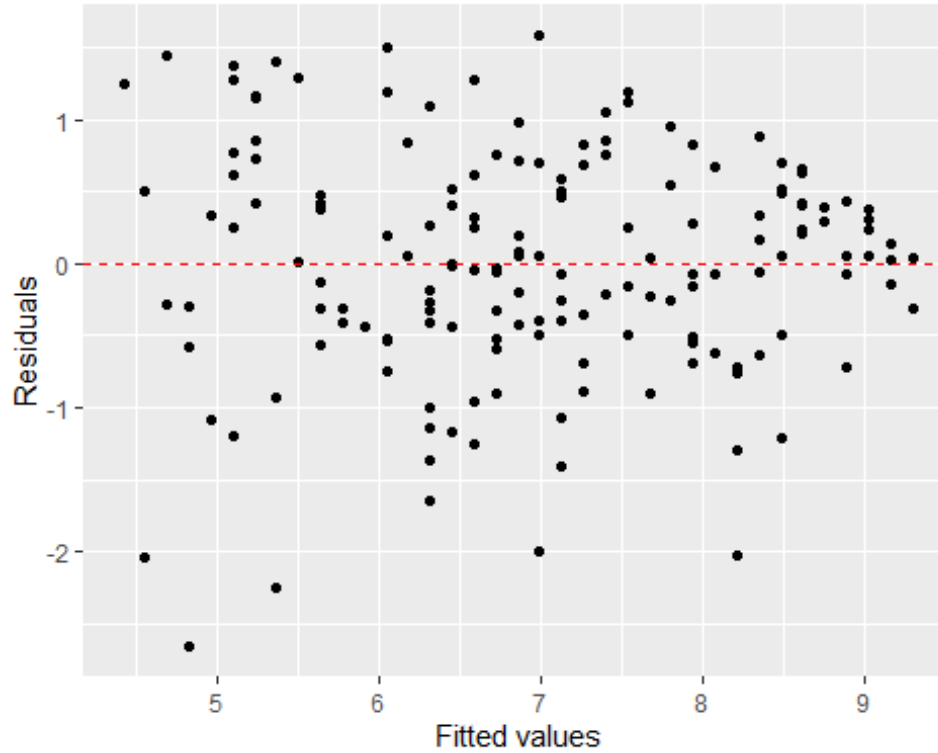
```
m1_aug <- augment(m1)
```

Q. If someone saw the least squares regression line and not the actual data, how would they predict a country's personal freedom score for one with a 3 rating for `pf_expression_control`? Is this an overestimate or an underestimate, and by how much? In other words, what is the residual for this prediction?

```
m1_aug <- augment(m1)
```

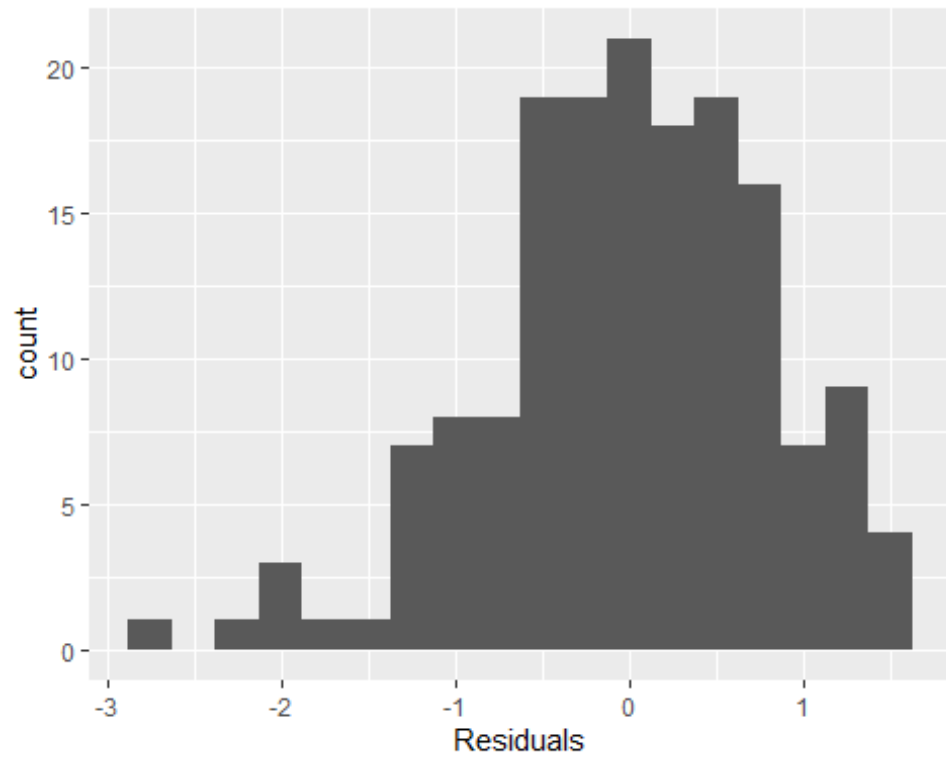
```
ggplot(data = m1_aug, aes(x = .fitted, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  xlab("Fitted values") +
```

```
ylab("Residuals")
```



##Is there any apparent pattern in the residuals plot? What does this indicate about the linearity of the relationship between the two variables?

```
ggplot(data = m1_aug, aes(x = .resid)) +  
  geom_histogram(binwidth = 0.25) +  
  xlab("Residuals")
```

##no the patterns

of the residuals look normal

##Q:1Based on the histogram, does the nearly normal residuals condition appear to be violated? Why or why not? ##A:the histogram looks normal and nearly normal residual conditions appear to meet